FACULTY OF ELECTRICAL ENGINEERING CTU IN PRAGUE

Thread biosensor bridge v1.0

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1 Overview

- 2 input signal 0-5V
- Lower treshold 0,45V
- Upper treshold adjustable
- EV e CV Cars
- Auto reset after 10s

2 Description

- The BSPD has the role of turning off the engine in the event of sudden braking and acceleration of more than 10% at the same time.
- It compliants the 2019 rules of FSG. The circuit works with active low logic 5V.
- The BRAKE COMPARATOR checks that the pressure in the brake circuit is within an interval (0 = error).
- The THROTTLE COMPARATOR checks that the throttle is open beyond a minimum threshold but less than 10% (0 = error).
- The output signals of the two comparators are in NOR (1 = error). The resistor R = 39 kΩ and the capacitor C = 10 uF operate a time constant of 390 ms.
- The TIME 1 COMPARATOR determines a delay of about 500ms (496ms) in the activation of the system (0 = error). The resistance R = 560 kΩ and the capacitor C = 10uF operate a time constant of 5600ms.
- The TIME 2 COMPARATOR determines a delay of about 9900ms after which the BSPD is deactivated (RESET CONDITION) and the engine can be restarted (0 = error). The NO RELAY performs the coupling with the car's powerline.
- Both the brake signal and the accelerator signal are compared with two thresholds, one upper and one lower.
- Lower threshold: 0.45V set via resistors R102 / R116 (100 Ω, pull-up) and R112 / R121 (10 kΩ, pull-down).
- Upper threshold: adjustable from 0.02 V to 5 V Set by a potentiometer M64X203KB40 R101 / R115 (10 Ω to 2.2 MΩ, pull-up) and a resistor (10 kΩ, pull-down).
- Diodes inversely polarized operate a protection against peaks on all the signals.
- Although in the board is written "BSPD v0.2", don't worry, the version produced is v0.3 avalaible on the official site of FSG.

3 Technical specification

	Unit	Value
Power voltage	V	13.8
Maximum current through SC	A	3
Dimensions	$mm * mm * mm$	44*36*11 (without cable)
Weight	g	13
Temperature range	C	-20 ÷ 60
n° of inputs	$--$	2
Voltage range signal input	V	0 ÷ 5